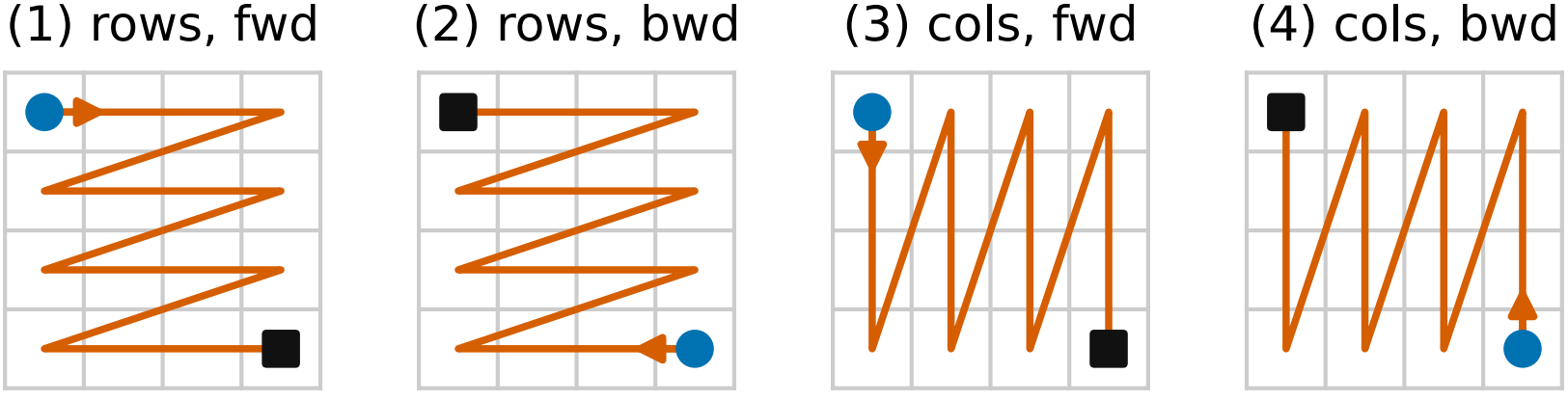
(1) rows, fwd
(2) rows, bwd
(3) cols, fwd
(4) cols, bwd